



Operating System Security (MITS 74915)

Submitted to,

Name: Julie Thorpe

Faculty of Business and Information Tech

Submitted by,

Name: Avishkar Sharma

Assignment 1

Table of Contents

Question. No. 1	1
Question. No. 1.a.(i&ii)	1
Question. No. 1(b).....	2
Question. No. 1(c).....	2
Question. No. 2	3
Question. No.2(a).....	3
Question. No. 2b.(i).....	3
Question. No. 2b.(ii).....	4

Question. No. 1

Question. No. 1.a.(i&ii)

```
root@fedora:/home# id Sarah AvishkarSharma
uid=1001(Sarah_AvishkarSharma) gid=1001(Sarah_AvishkarSharma) groups=1001(Sarah_AvishkarSharma)
root@fedora:/home# id Evelyn AvishkarSharma
uid=1002(Evelyn_AvishkarSharma) gid=1002(Evelyn_AvishkarSharma) groups=1002(Evelyn_AvishkarSharma)
root@fedora:/home# ls
avishkar Evelyn_AvishkarSharma Sarah_AvishkarSharma
root@fedora:/home# chmod 701 Sarah_AvishkarSharma
root@fedora:/home# su Sarah_AvishkarSharma
Sarah_AvishkarSharma@fedora:/home$ cd Sarah_AvishkarSharma
Sarah_AvishkarSharma@fedora:~$ pwd
/home/Sarah_AvishkarSharma
Sarah_AvishkarSharma@fedora:~$ echo "This is a Secret File" > /home/Sarah_AvishkarSharma/codes.txt
Sarah_AvishkarSharma@fedora:~$ chmod 604 /home/Sarah_AvishkarSharma/codes.txt
Sarah_AvishkarSharma@fedora:~$ tree -pug /home/Sarah_AvishkarSharma
[drwx-----x Sarah_AvishkarSharma Sarah_AvishkarSharma] /home/Sarah_AvishkarSharma
└─ [-rw----r-- Sarah_AvishkarSharma Sarah_AvishkarSharma] codes.txt

1 directory, 1 file
Sarah_AvishkarSharma@fedora:~$
```

Image 1: User Sarah ensuring user Evelyn can't access.

```
Evelyn_AvishkarSharma@fedora:~$ ls /home/Sarah_AvishkarSharma/
ls: cannot open directory '/home/Sarah_AvishkarSharma/': Permission denied
```

Image 2: Verification from user Evelyn

The image above shows that assigning permission was successful in blocking user Evelyn_AvishkarSharma from performing ls its contents; hence, Sarah_AvishkarSharma is safe from harm. Since the parent directory now acts as a barrier.

Question. No. 1(b)

```
root@localhost-live:/home/liveuser# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0       7:0    0   2.4G  1 loop /run/rootfsbase
sda         8:0    0   15G   0 disk
├─sda1      8:1    0    1M   0 part
├─sda2      8:2    0    2G   0 part
└─sda3      8:3    0   13G   0 part
sr0         11:0    1   2.7G  0 rom  /run/initramfs/live
zram0       251:0   0   2.9G  0 disk [SWAP]
root@localhost-live:/home/liveuser# mkdir /mnt/Sarah_fedora
root@localhost-live:/home/liveuser# mount /dev/sda3 /mnt/Sarah_fedora/
root@localhost-live:/home/liveuser# cat /mnt/Sarah_fedora/root/etc/pass
passim.conf  passwd      passwd-     passwdqc.conf
root@localhost-live:/home/liveuser# cat /mnt/Sarah_fedora/root/etc/passwd | grep Sarah
Sarah_AvishkarSharma:x:1001:1001::/home/Sarah_AvishkarSharma:/bin/bash
root@localhost-live:/home/liveuser# tree -puq /mnt/Sarah_fedora/home/Sarah_AvishkarSharma/
[drwx----x 1001 1001 ] /mnt/Sarah_fedora/home/Sarah_AvishkarSharma/
└─ [-rw---r-- 1001 1001 ] codes.txt

1 directory, 1 file
root@localhost-live:/home/liveuser# cat /mnt/Sarah_fedora/home/Sarah_AvishkarSharma/codes.txt
This is a Secret File
root@localhost-live:/home/liveuser# cat /mnt/Sarah_fedora/root/etc/passwd | grep 1001
Sarah_AvishkarSharma:x:1001:1001::/home/Sarah_AvishkarSharma:/bin/bash
root@localhost-live:/home/liveuser#
```

Figure 3: Booting into fedora by Evelyn.

User Evelyn_AvishkarSharma boots into a Live OS using fedora ISO. In this way, it uses the live OS account which has full administrative control over hardware and can't recognize the original Group ID of 1001 restrictions to original OS where it saw using "lsblk", enabling to mount partition *sda3* and bypasses the software permissions set by the original Fedora OS. Above screenshot shows that Live OS doesn't enforce Sarah_AvishkarSharma directory restriction (drwx----x), thereby allowing Evelyn to read the contents of codes.txt right off the file system.

Question. No. 1(c)

Encryption of Sarah's information will maintain its privacy for sure, but the lack of permission settings allows a "File Replacement" attack. In Linux, the parent directory holds permission to delete or rename a file not the file itself. This implies that an attacker could easily delete an encrypted file that belongs to Sarah and can replace it with a same name but hostile or corrupted. Subsequently, while trying to decrypt or work with those files, she would be unknowingly dealing with attacker's substituted data.

Question. No. 2

Question. No.2(a)

In this case, the aim is to affect persistence. This is a method attackers use to maintain control over a system even if it is restarted or logged out. One way to do so is by editing the Windows Registry, and much more specifically, its Run key.

In effect, a harmful .exe path is added to the registry key:

HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run. The OS will consequently run this file each time the user logs in or even logs off. Therefore, after the user logs back in or out, the "malware" can start running on its own without any human interaction.

Reference: <https://attack.mitre.org/techniques/T1547/001/>

Question. No. 2b.(i)

- Operation contains RegSetValue (Include) – It captures when an attacker creates or modifies a registry value in the Run key.
- Operation is RegCreateKey (Include) – It captures if the attacker needs to create the Run key first.
- Operation is RegDeleteValue (Include) – It captures if anyone deletes anything in the Run directory.
- Result is SUCCESS (Include) – It focuses only on successful modifications, eliminating noise from failed attempts.
- Path Contains CurrentVersion\Run and Included, to target the registry location for auto start persistence.
- Process Name is AvishkarSharma (Exclude) – excludes my own processes as required.
- Process Name is sihost.exe (Exclude) – This was running in background, so to view registry clearly, I excluded this.

Question. No. 2b.(ii)

Time of Day	Process Name	PID	Operation	Path	Result	Detail
11:31:35.3902486 PM	regedit.exe	25348	RegDeleteValue	HKCU\Software\Microsoft\Windows\CurrentVersion\Run\Assignment	SUCCESS	
11:31:43.4521476 PM	regedit.exe	25348	RegSetValue	HKCU\Software\Microsoft\Windows\CurrentVersion\Run\New Value #1	SUCCESS	Type: REG_SZ Length: 2 Data:
11:31:48.1393678 PM	regedit.exe	25348	RegSetValue	HKCU\Software\Microsoft\Windows\CurrentVersion\Run\Assignment	SUCCESS	Type: REG_SZ Length: 2 Data:
11:31:48.1395875 PM	regedit.exe	25348	RegDeleteValue	HKCU\Software\Microsoft\Windows\CurrentVersion\Run\New Value #1	SUCCESS	
11:32:01.9994684 PM	regedit.exe	25348	RegSetValue	HKCU\Software\Microsoft\Windows\CurrentVersion\Run\Assignment	SUCCESS	Type: REG_SZ Length: 28 Data: "notepad.exe"

Column	Relation	Value	Action
☒ Operation	contains	RegSetValue	Include
☒ Operation	is	RegDeleteValue	Include
☒ Operation	is	RegCreateKey	Include
☒ Result	is	SUCCESS	Include
☒ Path	contains	CurrentVersion\Run	Include
☒ Process Name	is	AvishkarSharma	Exclude
☒ Process Name	is	sihost.exe	Exclude

Figure 4: The Process Monitor filter and RegSetValue operation.

As seen in Figure 1, which resulted in thousands of background noise events ranging from pings to the usual updates that the system does. However, by applying the required targeted filters, I successfully isolated the relevant registry modifications with PID 25348.